

EXAMPLE OPERATIONAL WORKFLOWS AND USE SCENARIOS

Purpose of Workflow Disclosure

The following operational workflows are provided as illustrative examples demonstrating how the longitudinal penile physiology monitoring system may function during real-world use. These workflows are non-limiting and are intended to support a broad range of device implementations, computational architectures, and user interaction models.

Workflow Example 1 — Baseline Monitoring

A user may wear the device during normal daily activity or sleep. The device may periodically sample mechanical deformation, temperature, motion context, and optional perfusion signals. Collected data may be stored locally or transmitted to a computational platform where baseline physiologic patterns are established and updated over time.

Workflow Example 2 — Nocturnal Physiology Assessment

During sleep, the device may operate in a low-power monitoring mode detecting nocturnal physiologic activation events. Event detection algorithms may identify onset timing, duration, rigidity stability, and recovery behavior while contextual information from motion or systemic monitoring devices may distinguish sleep stages or movement artifacts.

Workflow Example 3 — Structural Calibration Integration

Structural geometry data obtained from a companion scanning device may be referenced to interpret real-time measurements relative to individualized anatomical capacity. The device may classify activation state based on proportional engagement rather than absolute deformation.

Workflow Example 4 — Context-Aware Physiologic Interpretation

Data obtained from systemic monitoring devices may be synchronized with localized measurements to determine whether detected physiologic activation corresponds to exercise, stress response, sleep transitions, or other contextual influences. Context-aware modeling may refine analytic outputs and reduce false classification.

Workflow Example 5 — Multi-Node Unified Analysis

Multiple nodes may transmit synchronized datasets to a computational system performing cross-node analysis. Derived metrics, adaptive indices, or abstract profiles may be generated and presented to the user through privacy-preserving visualization methods.

Workflow Example 6 — Longitudinal Adaptation and Learning

Over extended monitoring periods, adaptive algorithms may refine baseline models, adjust sensing parameters, and improve predictive interpretation of physiologic trends. Longitudinal learning may enable personalized analytics, coaching feedback, or compatibility analysis.

The workflows described herein are illustrative and non-limiting and encompass any sequence of operations enabling wearable longitudinal physiologic monitoring and multi-node data interpretation.